M201103W- Signals and Systems

ssch7_1 –Representation of D-T signals in the z-domain(Similar to the s-domain)

14:10 pm Wednesday @ C107, 1st Teaching Building

14:10 pm Friday @ A107, 1st Teaching Building

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Question

A D-T LTI system is described by the difference equation
$y[k] + 3\,y[k-1] + 2\,y[k-2] = x[k]$ with initial conditions: $y[-1] = 0$, $y[-2] = 0.5$.
Given $x[k] = u[k]$. Analyse the following items.

※ Calculate the response of the system.

In the time domain:
$y[k] = y_{zi}[k] + y_{zs}[k]$ $(x[k]*h[k])$

In the frequency domain(DTFT):
$Y(e^{j\Omega}) = X(e^{j\Omega})H(e^{j\Omega})$, only with 0 initial state, and input $x[k]$ is absolutely summable. The system should be stable.

※ Analyse the performance of the system. How to deal with those disadvantages?

In the time domain:
Described by $h[k]$.

In the frequency domain:
Described by $H(e^{j\Omega})$.
Only suitable for stable systems.

Analysis of D-T signals and systems in the z-domain(similar to the s-domain)

Analysis of D-T signals in the z-domain

- Representation of signals in the z-domain
- z-Transform of elementary unilateral signals.
- Properties of unilateral z-Transform
- Inverse z-Transform

Analysis of D-T systems in the z-domain

- Representation of systems
- System function and system performance
- Response in the z-domain
- Block diagram of system

1. Expansion of DTFT: from $X(e^{j\Omega})$ to $X(z)$

From DTFT to z-Transform

$$DTFT\{x[k]\} = X(e^{j\Omega}) = \sum_{k=-\infty}^{+\infty} x[k]e^{-j\Omega k} \qquad x[k] \rightarrow x[k]r^{-k}$$

$$DTFT\{x[k]r^{-k}\} = \sum_{k=-\infty}^{+\infty} x[k]r^{-k}e^{-j\Omega k}$$

$$= \sum_{k=-\infty}^{+\infty} x[k](re^{j\Omega})^{-k} \qquad \text{Let} \quad z = re^{j\Omega}$$

$$X(z) = \sum_{k=-\infty}^{+\infty} x[k]z^{-k}$$

Bilateral z-transform

Representation of D-T signals in the z-Transform

- z-Transform of D-T signals
- Definition of unilateral z-Transform
- ROC of z-Transform

Representation of D-T signals in the z-Transform

z-Transform representation of D-T signals: $x[k]$ can be represented as linear combinations of $\boldsymbol{z^k}$, only with different weighting coefficients $X(z)$.

$$x[k] = \frac{1}{2\pi \mathrm{j}} \oint_c X(z) z^{k-1} \mathrm{d}z$$

Where C is a closed curve in the ROC of $X(z)$.

※Bilateral z-Transform $$X(z) = \sum_{k=0}^{+\infty} x[k]\, z^{-k}$$

※Inverse bilateral z-Transform $$x[k] = \frac{1}{2\pi \mathrm{j}} \oint_c X(z) z^{k-1} \mathrm{d}z$$

2. Definition of unilateral *z*-Transform

※Unilateral *z*-Transform $X(z)=\sum_{k=0}^{+\infty} x[k]\, z^{-k}$

※Inverse unilateral z-Transform

$$x[k]=\frac{1}{2\pi \mathrm{j}}\oint_{c} X(z) z^{k-1}\mathrm{d}z$$

Where C is a closed curve in the ROC of $X(z)$.

※Symbolically: $x[k] \xleftrightarrow{\mathscr{Z}} X(z)$

z-Transform:

$X(z)=\mathscr{Z}\{x[k]\}$

Inverse z-Transform:

$x[k]=\mathscr{Z}^{-1}\{X(z)\}$

In sampling view, from s-Transform to z-Transform

s-Transform of ideal sampling signals:

$$x_s(t) = x(t)\sum_{k=-\infty}^{\infty}\delta(t-kT) = \sum_{k=-\infty}^{\infty}x(kT)\delta(t-kT)$$

s-Transform

$$X_s(s) = \mathscr{L}\{x_s(t)\} = \sum_{k=-\infty}^{+\infty}x(kT)\mathrm{e}^{-ksT}$$

Let $\boxed{z = \mathrm{e}^{sT},\ x[k] = x(kT)}$

$$\mathscr{L}\{x_s(t)\} = \sum_{k=-\infty}^{\infty}x[k]z^{-k} = X(z)$$

$$X_s(s) = \sum_{k=-\infty}^{+\infty}x(kT)\mathrm{e}^{-ksT}$$

$$X(z) = \sum_{k=-\infty}^{+\infty}x[k]z^{-k}$$

Mapping relation from s to z:

$$z = \mathrm{e}^{sT}$$

3. ROC of z-Transform

ROC $$\sum_{k=-\infty}^{+\infty} \left| x[k] z^{-k} \right| < \infty$$

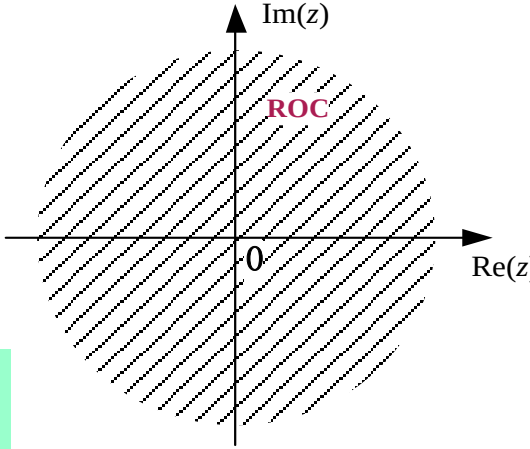
- Finite-length signals
- Right-sided signals (unilateral)
- Left-sided signals
- Bilateral signals

(1) Finite-length signals

$$X(z) = \sum_{k=N_1}^{N_2} x[k] z^{-k}$$

ROC $\quad 0 < |z| < \infty$

Example. $x[k] = \begin{cases} 1, & 0 \le k \le N-1 \\ 0, & \text{other} \end{cases} = R_N[k]$



$$X(z) = \sum_{k=0}^{N-1} z^{-k} = \frac{1 - z^{-N}}{1 - z^{-1}}$$

$|z| > 0$

(2) Right-sided signals

$$X(z) = \sum_{k=N_1}^{\infty} x[k]z^{-k}$$ ROC $|z| > R_-$

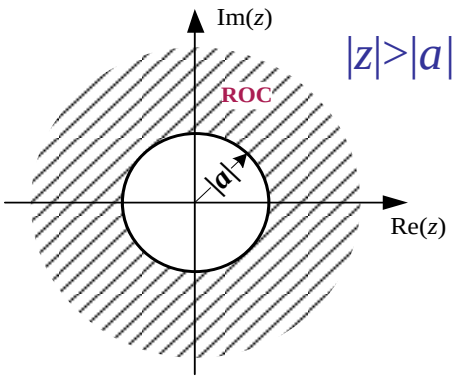
Example. Calculate unilateral z-Transform of $x[k] = a^k u[k]$

Determine the ROC.

Solution:

$$X(z) = \sum_{k=0}^{\infty} x[k]z^{-k} = \sum_{k=0}^{\infty} a^k z^{-k}$$

$$= \sum_{k=0}^{\infty} (az^{-1})^k = \frac{1}{1-az^{-1}}, \quad |z| > |a|$$



When $|z|>|a|$, $X(z)$ exists.

(3) Left-sided signals

$$X(z) = \sum_{k=-\infty}^{N_2} x[k] z^{-k} \quad \text{ROC} \quad |z| < R_+$$

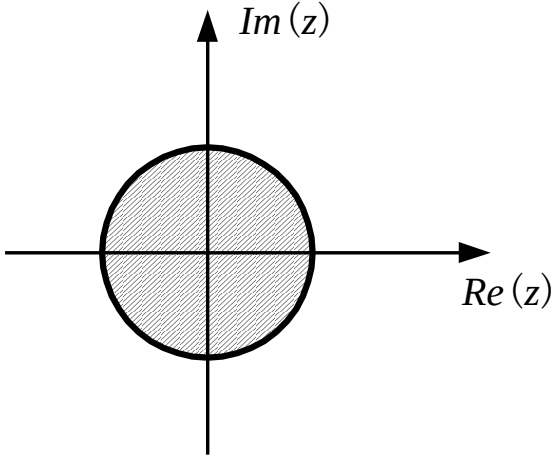
Example. Calculate the z-Transform of $x[k] = -b^k u[-k-1]$

Determine the ROC.

Solution: $$X(z) = \sum_{k=-\infty}^{-1} -b^k z^{-k} = \sum_{k=1}^{\infty} -b^{-k} z^k$$

$$= 1 - \sum_{k=0}^{\infty} b^{-k} z^k = 1 - \frac{1}{1 - b^{-1} z}$$

$$= \frac{1}{1 - b z^{-1}} = \frac{z}{z - b} \quad |z| < |b|$$



(4) Bilateral signals

$$X(z) = \sum_{k=-\infty}^{\infty} x[k] z^{-k}$$

ROC $R_- < |z| < R_+$

Example. Calculate unilateral z-Transform of $x[k]$. Determine the ROC.

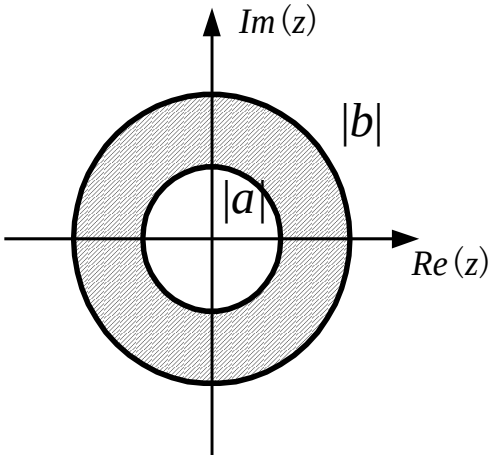
$$x[k] = a^k u[k] - b^k u[-k-1]$$

Solution: $$X(z) = \frac{1}{1-az^{-1}} + \frac{1}{1-bz^{-1}}$$

$$= \frac{z}{z-a} + \frac{z}{z-b}$$

Note: $|a| < |z| < |b|$

Only when $|b|>|a|$, $X(z)$ exists.



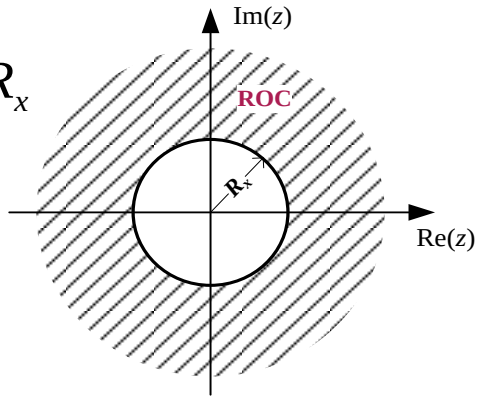
Summary

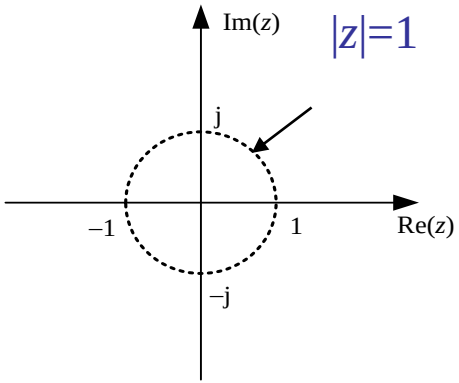
※Unilateral z-Transform: $$X(z) = \sum_{k=0}^{\infty} x[k] z^{-k}$$

$X(z)$ is power series of z^{-1}

ROC: z range that makes $X(z)$ above converge.

$|z| > R_x$





$|z|=1$ → z-Transform becomes DTFT.

ROC of z-Transform of a right-sided signal is outside of a certain circle area.